MountainCar AI: Theory and Results

Project scope: Person 1 implemented the reinforcement-learning environment, agent algorithms, experiments, and evaluation workflow for Gymnasium MountainCar-v0.

The state is position and velocity. The three actions push left, do nothing, or push right. Each step receives reward -1, so a higher return means the goal was reached in fewer steps. Episodes end at the goal or after 200 steps.

AI assistance was used for implementation support and testing. The project was reviewed and the reported experiments were executed locally.

Algorithms

Value iteration applies repeated Bellman optimality updates to a known finite transition model. The implementation is generic and is tested with a small finite MDP.

For the discounted-to-undiscounted MDP construction, continuation transitions use gamma times their original probability, continuation rewards use R/gamma, and the missing 1-gamma probability goes to an absorbing OUT state with zero reward. Substitution gives the original discounted Bellman equation.

Model-based Monte Carlo estimates transition counts and mean rewards before running value iteration. Tabular Q-learning learns directly from samples. Function approximation uses cosine Fourier features and linear weights. Constrained Q-learning removes actions predicted to exceed the velocity limit.

Experiment configuration

Seed: 42. Episodes: 1000 for the main comparisons. The model-based run is capped at 300 episodes because it performs a value-iteration pass over the collected model.

The tabular agent uses a 40 x 40 discretized state space, epsilon-greedy exploration, learning rate 0.1, and discount factor 0.99. Evaluation uses greedy actions with exploration disabled.

Measured results

tabular_q_learning: mean return=-200.00, goal rate=0.0%

function_approximation: mean return=-156.87, goal rate=83.0%

constrained_q_learning: mean return=-200.00, goal rate=0.0%

model_based_monte_carlo: mean return=-200.00, goal rate=0.0%

Less negative returns are better because every environment step has reward -1.